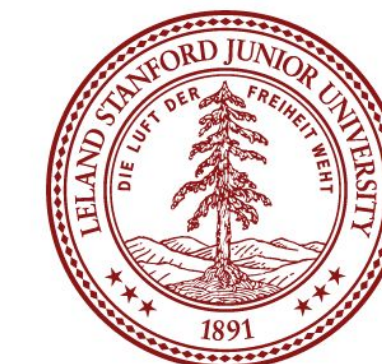




TurnOne: The Cost of Convention in Low-Rank Games

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CS234: Reinforcement Learning – Winter 2026

Problem

Expert Pokémon VGC strategies are maximally different from Nash equilibria (TV = 0.99), yet expert-vs-expert payoffs match Nash within 0.02. Why does convention work in a ~200-action zero-sum game, and can offline RL beat it?

Pokémon VGC is a competitive doubles format where each player builds a team of six Pokémon, brings four to battle, and selects two as leads. On turn 1, both players simultaneously choose actions for their two active Pokémon. Each Pokémon can pick one of four moves targeted at either opponent or its partner, and the player may terastallize one Pokémon. This produces a combinatorial action space of ~200 valid combinations per side.

We formalize turn-1 play as an offline contextual bandit: context is the full game state, action is one of ~200 valid turn-1 combinations, and reward comes from a learned dynamics model. The game is zero-sum with payoffs derived from HP differential, KOs, and field state.

Data

We use **154,718 rated battles** from Pokémon Showdown (Gen 9 VGC Regulation G, 1500+ Elo), **yielding 309,436 directed turn-1 examples**. Each encodes team composition (6 Pokémon with moves, abilities, items), chosen leads, and field state.

Approach

Reward model: A dynamics model (shared encoder + action embeddings) predicts per-Pokémon HP changes, KO probabilities, and post-turn field state, defining the bandit reward $r(s, a, a')$. HP MAE = **13.3/100**, KO AUC = **0.91**.

Three strategies evaluated on this reward model:

Behavior Cloning imitates expert play, ignores reward.

Transformer encoder (4 layers, 4 heads, $d=128$), 12 Pokémon tokens + 1 field token, independent action and tera heads. 44.8% top-1 (3× uniform), 79.6% top-3.

Nash computed via LP on the dynamics model's payoff matrix.

Conservative Q-Learning is the offline RL baseline. Factored Q-network $Q(s, a) = Q_a + Q_b + Q_{\text{tera}}$ with the same encoder. Penalizes OOD Q-values. At horizon 1, reduces to conservative reward regression with no bootstrapping.

The puzzle: exploitable experts, zero payoff gap

Experts are exploitable by 1.41 reward units (about half a KO per turn) and play nothing like Nash (TV = 0.99).

Nash puts mass on ~2.7 action vs BC spreads across ~95 actions

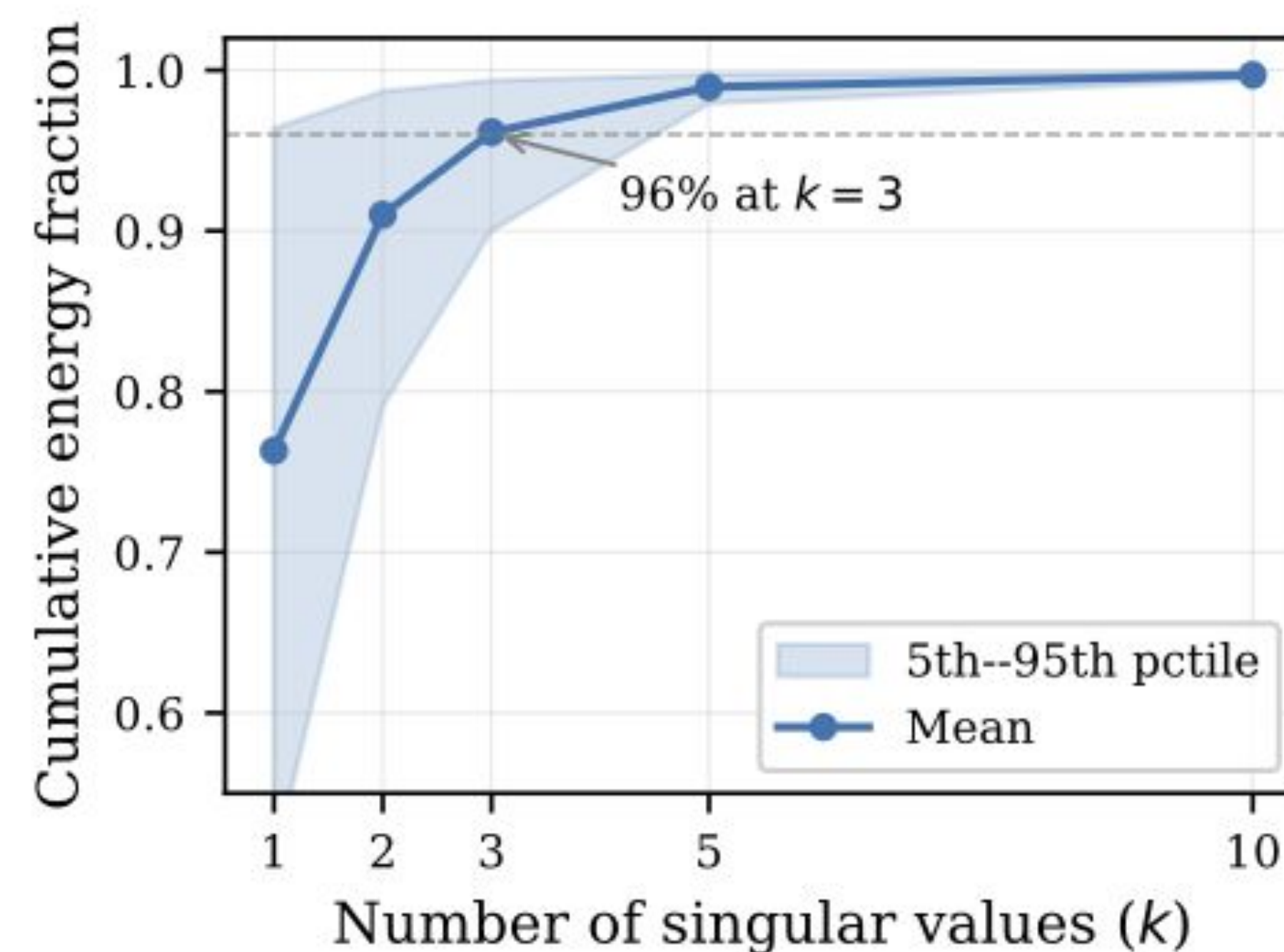
Table 1. Payoff landscape for conventional vs. optimized play (500 matchups, mean values).

Scenario	Payoff (P1 perspective)
BC worst-case	-1.19
Nash value V^*	+0.22
BC-vs-BC	+0.20
Best-response to BC	+1.63
Exploitability	1.41 (vs. optimizer)
Cross-play gap	-0.02 (vs. convention)

Why convention is free: the game is 3-dimensional

The payoff matrix has effective rank 3 out of ~122 actions. Top 3 singular values capture 96% of spectral energy. Most of the ~200 nominal actions are functional duplicates that differ in name but produce similar outcomes.

- 93% of BC–Nash deviation energy sits in the payoff null space
- Three effective dimensions: type advantage, raw damage, speed/priority
- Interaction decomposition: $\Delta_1 = +1.03$, $\Delta_2 = -1.02$, cross-term = -0.03 . Deviations are orthogonal to R's column/row spaces, so they cancel.

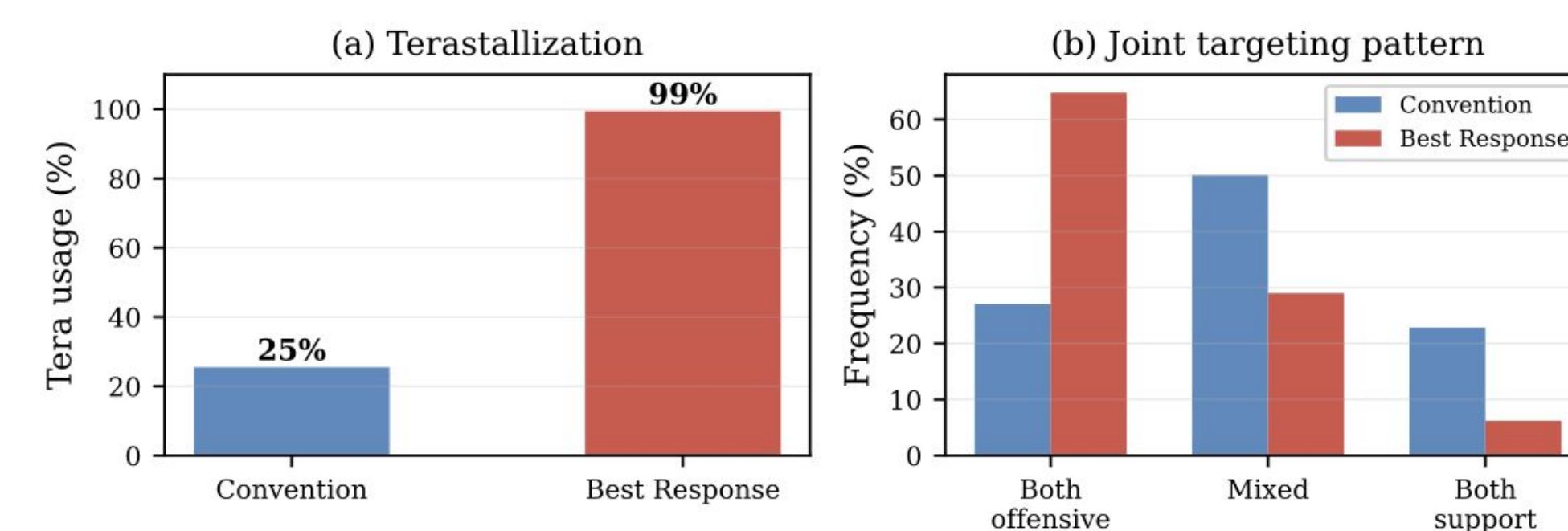


Where convention fails: terastallization

Tera is irreversible: use it once and it's gone for the game.

Experts conserve it for later turns (25% usage on turn 1), but myopic best response says use it now (99%).

- Tera alone accounts for 30% of total exploitability (0.41/1.41)
- Best responses also shift targeting: 65% both-offensive vs. 27% for BC



CQL: reward optimization can't escape convention

Switching from BC to CQL reshapes 60% of the action

distribution but reduces exploitability by only 17%. Of the

CQL–BC distributional shift, 94% lands in the payoff null space.

Table 5. BC vs. CQL vs. Nash (500 matchups, 95% CI).

Metric	BC	CQL	Nash
Exploitability	1.40 [1.36, 1.46]	1.15 [1.08, 1.23]	0.00
TV(·, Nash)	0.99	0.98	0.00
Worst-case value	-1.23	-0.99	+0.16

Conclusions

Payoff matrices have effective rank 3. 93% of strategic variation is payoff-irrelevant; most actions are functional duplicates.

The exception: terastallization, a one time resource, accounts for 30% of exploitability. Irreversibility makes convention costly.

CQL confirms the bottleneck is game structure. Reward optimization reshuffles the null space without escaping it.

Prediction: convention should be free in games with low effective rank (RTS, sports playcalling) and costly where irreversible commitments dominate (poker all-ins, auctions).